

Comparison Dimensions	Stage I: Modeling & Opt.	Stage II: Real-time Control	Stage III: Data-driven
Key Methods	MPC, Mixed-integer Opt., Geometric Leash Models	Impedance/Admittance, Force Observers, CBF	RL, BC, TCN-MPC, Diffusion Policy, RMA
Time Scale	Long-term (Finite Horizon)	Short-term (Instantaneous)	Variable (Adaptive)
Computational Cost	High (Nonlinear Opt.)	Low (Feedback Control)	Medium (Inference)
Core Advantage	Physical Interpretability	Real-time Robustness	Complex Adaptability
Modeling Root	First-principles Dynamics	Simplified Interaction Laws	Neural Representation